

Ex. No. : 04

Date:

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A PYTHON PROGRAM TO IMPLEMENT SINGLE LAYER PERCEPTRON

Aim:

To implement a **Single Layer Perceptron model in Python** to classify **Diabetes dataset**.

Requirements:

Jupyter Notebook with Python, Diabetes.csv dataset, and required Python libraries.

PROGRAM:

Step 1: Import Required Libraries

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
from sklearn.preprocessing import StandardScaler
```

Step 2: Upload Dataset

```
from
sklearn.datasets

import

load_diabetes
```

Step 3: Load the Dataset

```
data = load_diabetes(as_frame=True).frame
data.head()
```

	age	sex	bmi	bp	s1	s2	s3	s4	s5	s6	target
0	0.038076	0.050680	0.061696	0.021872	-0.044223	-0.034821	-0.043401	-0.002592	0.019907	-0.017646	151.0
1	-0.001882	-0.044642	-0.051474	-0.026328	-0.008449	-0.019163	0.074412	-0.039493	-0.068332	-0.092204	75.0
2	0.085299	0.050680	0.044451	-0.005670	-0.045599	-0.034194	-0.032356	-0.002592	0.002861	-0.025930	141.0
3	-0.089063	-0.044642	-0.011595	-0.036656	0.012191	0.024991	-0.036038	0.034309	0.022688	-0.009362	206.0
4	0.005383	-0.044642	-0.036385	0.021872	0.003935	0.015596	0.008142	-0.002592	-0.031988	-0.046641	135.0

Step 4: Define Features and Target

Target column:

```
data['Class'] = (data['target'] > data['target'].mean()).astype(int)
```

```
X = data.drop(["target", "Class"], axis=1)
y = data["Class"]
```

Step 5: Feature Scaling

Perceptron works better when data is scaled.

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
```

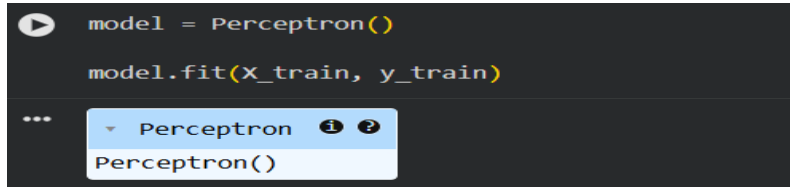
Step 6: Split Dataset

```
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)
```

Step 7: Train the Perceptron Model

```
model = Perceptron()
```

```
model.fit(X_train, y_train)
```



Step 8: Predict the Results

```
y_pred = model.predict(X_test)
```

Step 9: Evaluate the Model

Accuracy

```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.7303370786516854
```

Confusion Matrix

```
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

```
[[ 32  18]
 [  6  33]]
```

Classification Report

```
print("Classification Report:")
print(classification_report(y_test, y_pred))
```

Classification Report:					
	precision	recall	f1-score	support	
0	0.84	0.64	0.73	50	
1	0.65	0.85	0.73	39	
accuracy			0.73	89	
macro avg	0.74	0.74	0.73	89	
weighted avg	0.76	0.73	0.73	89	

Result:

Thus, the **Single Layer Perceptron model** was successfully implemented using Python and evaluated using the Diabetes dataset.